Aditya Rana

Robotics & Control Engineer

RESUME_1730

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Total Experience: 4 years

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 4 years of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Simulink, Gazebo, SLAM, Embedded Systems, PLC, Python, MATLAB, C++, PID Control, ROS 2, Motion Planning, OpenCV

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Robotics & Control Engineer | DeltaPoint Innovations | Jan 2024 - Present

- Lowered navigation recovery events by 16% using improved obstacle handling
- Improved pick-position repeatability by 39% after calibration updates
- Reduced robotic path completion time by 14% through route and controller tuning
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Robotics & Control Engineer | KoraLink Innovations | Jul 2022 - Dec 2023

- Reduced robotic path completion time by 23% through route and controller tuning
- Validated 316 robot simulation scenarios with recorded sensor logs

PROJECTS

Warehouse Mapping Trial | Embedded Systems, ROS 2, PLC, Python

- Benchmarked LiDAR SLAM configurations under moving obstacles.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Vision-Guided Pick Cell | MATLAB, ROS 2, SLAM, Gazebo

- Integrated camera calibration, pick pose estimation and manipulator trajectories.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Pragati Institute of Applied Sciences, 2022

Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

Industrial Robot Safety (2023)